

Datasets are found in the github repo <https://github.com/sumsar027/Master-NTNU->  
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Loading all packages needed for our programming

Saving all datasets from Refinitiv

```
# saving file path for all datasets

base_path <- "C:/Users/Eier/Desktop/Master balance sheet/balancesheets2"

# saving all names for bank files

bank_files <- c(

  # market based
  gs = "goldmansachs.xlsx",
  ms = "morganstanley.xlsx",

  # universal/commercial
  jpm = "jpmorgan.xlsx",
  boa = "bankofamerica.xlsx",
  cg = "citigroup.xlsx",
  wf = "wellsfargo.xlsx",

  # custodian
  bny = "bny (1).xlsx",
  ss = "statestreet (1).xlsx"
)

# making a function for reading all datasets

for (bank in names(bank_files)) {
  file_path <- file.path(base_path, bank_files[bank])

  # assigning names for each bank and each excel sheet
  assign(paste0(bank, "_bs"), read_excel(file_path, sheet = 1))
  assign(paste0(bank, "_is"), read_excel(file_path, sheet = 2))
  assign(paste0(bank, "_fs"), read_excel(file_path, sheet = 3))
}

# saving the var dataset

var_data <- read_csv("C:\\Users\\Eier\\Desktop\\Master balance sheet\\balancesheets2\\var_99
(1).csv")
```

Data manipulation

```

# these are the bs rows we want to keep
keep_bs <- c(
  "Field Name",
  "Total Assets",
  "Collateralized Financing Agreements, REPO Financing Loans - Liabilities",
  "Total Liabilities",
  "Total Shareholders' Equity - including Minority Interest & Hybrid Debt",
  "Risk Weighted Assets",
  "Liquidity Coverage Ratio - Basel 3 - %",
  "Leverage Ratio - Basel 3 - %"
)

# these are the fs rows we want to keep
keep_fs <- c(
  "Field Name",
  "Capital Adequacy - Core Tier 1 (%)",
  "Dividend Payout Ratio - %",
  "Return on Average Total Assets - % (Income before Discontinued Operations & Extraordinary Items), TTM"
)

# this just cleans text a bit so matching works better
clean_labels <- function(x) {
  x %>%
    as.character() %>%
    str_squish()
}

# this keeps the first real value if there are duplicates
# so if one value is NA and one is a number, it keeps the number
pick_value <- function(x) {
  x <- as.character(x)
  x <- trimws(x)
  x[x %in% c("NA", "", "NaN", "NULL")] <- NA
  x <- x[!is.na(x)]

  if (length(x) == 0) {
    NA_character_
  } else {
    x[1]
  }
}

# this makes sure quarter is read as a real date
# it handles mixed date formats a bit more safely
parse_quarter <- function(x) {
  x <- as.character(x)
  x <- trimws(x)

  # this removes time if the value looks like a datetime
  x <- sub(" .*", "", x)

  # this turns empty strings into NA
  x[x %in% c("", "NA", "NULL", "NaN")] <- NA

```

```

# this tries a few different date formats
out <- suppressWarnings(lubridate::ymd(x))

missing <- is.na(out)
out[missing] <- suppressWarnings(lubridate::dmy(x[missing]))

missing <- is.na(out)
out[missing] <- suppressWarnings(lubridate::mdy(x[missing]))

as.Date(out)
}

# this cleans one sheet and reshapes it
clean_sheet <- function(df, keep_rows) {

  # this grabs the first column name
  first_col <- names(df)[1]

  # this renames the first column so it is easier to work with
  df <- df %>%
    rename(field_name = all_of(first_col)) %>%
    mutate(field_name = clean_labels(field_name))

  # this keeps only the rows we care about
  df_keep <- df %>%
    filter(field_name %in% keep_rows)

  # this saves the original quarter columns
  quarter_cols <- names(df_keep)[-1]

  # this pulls out the date row directly from the sheet
  quarter_vec <- df_keep %>%
    filter(field_name == "Field Name") %>%
    slice(1) %>%
    select(-field_name) %>%
    unlist(use.names = FALSE)

  # this turns quarter into a proper date column
  quarter_df <- tibble(
    col_id = quarter_cols,
    quarter = parse_quarter(quarter_vec)
  )

  # this removes the date row so only financial variables are left
  value_rows <- df_keep %>%
    filter(field_name != "Field Name")

  # this collapses duplicate row names before reshaping
  value_rows <- value_rows %>%
    group_by(field_name) %>%
    summarise(
      across(
        .cols = everything(),
        .fns = pick_value
      ),
      .groups = "drop"
    )

```

```

)

# this turns the data into long format
long_df <- value_rows %>%
  pivot_longer(
    cols = -field_name,
    names_to = "col_id",
    values_to = "value"
  )

# this joins the quarter column back in
# then it spreads the variables into regular columns
value_df <- long_df %>%
  left_join(quarter_df, by = "col_id") %>%
  pivot_wider(
    names_from = field_name,
    values_from = value
  ) %>%

# this moves quarter to the front
select(quarter, everything(), -col_id) %>%

# this renames columns to the short names we want
rename(
  total_assets = any_of("Total Assets"),
  repo = any_of("Collateralized Financing Agreements, REPO Financing Loans - Liabilities"),
  total_debt = any_of("Total Liabilities"),
  total_equity = any_of("Total Shareholders' Equity - including Minority Interest & Hybrid
Debt"),
  rwa = any_of("Risk Weighted Assets"),
  lcr = any_of("Liquidity Coverage Ratio - Basel 3 - %"),
  slr = any_of("Leverage Ratio - Basel 3 - %"),
  cet1 = any_of("Capital Adequacy - Core Tier 1 (%)"),
  dp = any_of("Dividend Payout Ratio - %"),
  roa = any_of("Return on Average Total Assets - % (Income before Discontinued Operations &
Extraordinary Items), TTM")
) %>%

# this turns all non-date columns into numeric
mutate(across(-quarter, as.numeric)) %>%

# this drops rows where quarter is missing
filter(!is.na(quarter)) %>%

# this keeps one row per quarter
distinct(quarter, .keep_all = TRUE) %>%

# this sorts from newest to oldest
arrange(desc(quarter))

value_df
}

# here we clean all bs datasets
gs_bs <- clean_sheet(gs_bs, keep_bs)
ms_bs <- clean_sheet(ms_bs, keep_bs)

```

```

jpm_bs <- clean_sheet(jpm_bs, keep_bs)
boa_bs <- clean_sheet(boa_bs, keep_bs)
cg_bs  <- clean_sheet(cg_bs, keep_bs)
wf_bs  <- clean_sheet(wf_bs, keep_bs)
bny_bs <- clean_sheet(bny_bs, keep_bs)
ss_bs  <- clean_sheet(ss_bs, keep_bs)

# here we clean all fs datasets
gs_fs <- clean_sheet(gs_fs, keep_fs)
ms_fs <- clean_sheet(ms_fs, keep_fs)
jpm_fs <- clean_sheet(jpm_fs, keep_fs)
boa_fs <- clean_sheet(boa_fs, keep_fs)
cg_fs <- clean_sheet(cg_fs, keep_fs)
wf_fs <- clean_sheet(wf_fs, keep_fs)
bny_fs <- clean_sheet(bny_fs, keep_fs)
ss_fs <- clean_sheet(ss_fs, keep_fs)

# here we merge bs and fs for each bank
gs <- left_join(gs_bs, gs_fs, by = "quarter")
ms <- left_join(ms_bs, ms_fs, by = "quarter")
jpm <- left_join(jpm_bs, jpm_fs, by = "quarter")
boa <- left_join(boa_bs, boa_fs, by = "quarter")
cg <- left_join(cg_bs, cg_fs, by = "quarter")
wf <- left_join(wf_bs, wf_fs, by = "quarter")
bny <- left_join(bny_bs, bny_fs, by = "quarter")
ss <- left_join(ss_bs, ss_fs, by = "quarter")

# here we sort again after merging just to be safe
gs <- gs %>% arrange(desc(quarter))
ms <- ms %>% arrange(desc(quarter))
jpm <- jpm %>% arrange(desc(quarter))
boa <- boa %>% arrange(desc(quarter))
cg <- cg %>% arrange(desc(quarter))
wf <- wf %>% arrange(desc(quarter))
bny <- bny %>% arrange(desc(quarter))
ss <- ss %>% arrange(desc(quarter))

# here we add the bank name and bank group
gs <- gs %>% mutate(bank = "gs", bank_group = "market")
ms <- ms %>% mutate(bank = "ms", bank_group = "market")
jpm <- jpm %>% mutate(bank = "jpm", bank_group = "universal")
boa <- boa %>% mutate(bank = "boa", bank_group = "universal")
cg <- cg %>% mutate(bank = "cg", bank_group = "universal")
wf <- wf %>% mutate(bank = "wf", bank_group = "universal")
bny <- bny %>% mutate(bank = "bny", bank_group = "custodian")
ss <- ss %>% mutate(bank = "ss", bank_group = "custodian")

# here we stack all bank datasets into one big dataset
bank_data <- bind_rows(
  gs,
  ms,
  jpm,
  boa,
  cg,
  wf,
  bny,

```

```
    ss
  ) %>%

  # sort the full dataset by bank and quarter
  arrange(bank, desc(quarter))
```

Adding VaR and VIX numbers to the finished dataset

```

# here we clean the bank names in the var dataset
var_data <- var_data %>%

# make sure the date column is a real date
mutate(date = as.Date(date)) %>%

# recode full bank names to the same short bank codes as in bank_data
mutate(
  bank = recode(
    bank,
    goldmansachs = "gs",
    morganstanley = "ms",
    jpmorgan = "jpm",
    bankofamerica = "boa",
    citigroup = "cg",
    wellsfargo = "wf",
    bny = "bny",
    statestreet = "ss"
  )
)

# here we merge var_data into bank_data
bank_data <- bank_data %>%

# join by bank and date
left_join(
  var_data,
  by = c("bank", "quarter" = "date")
)

# here we clean the vix dataset
vix_data <- read_csv("C:\\Users\\Eier\\Desktop\\Master balance sheet\\VIXCLS (1).csv") %>%

# make the date column a real date
mutate(observation_date = as.Date(observation_date)) %>%

# move the date to quarter end so it matches bank_data
mutate(
  quarter = ceiling_date(observation_date, unit = "quarter") - days(1)
) %>%

# rename the vix column
rename(vix = VIXCLS) %>%

# keep only what we need
select(quarter, vix)

# here we merge vix into bank_data
bank_data <- bank_data %>%

# join by quarter
left_join(vix_data, by = "quarter") %>%
# cutting out all data before 2010-03-31
filter(quarter >= as.Date("2010-03-31"))

```

## Descriptiv statistics



```

# descriptive statistics

# this makes a copy so the original bank_data does not get changed
desc_data <- bank_data %>%

# makes sure quarter is a date
mutate(
  quarter = as.Date(quarter)
) %>%

# sort by bank and quarter
arrange(bank, quarter) %>%

# calculate variables within each bank
group_by(bank) %>%
mutate(

  # Leverage measure
  leverage = total_assets / total_equity,

  # growth in Leverage
  leverage_growth_calc = leverage - lag(leverage, 1),

  # growth in var
  var_99_gaussian_growth = 100 * (log(var_99_gaussian) - log(lag(var_99_gaussian, 1))),

  # growth in balance sheet variables
  total_assets_growth = 100 * (log(total_assets) - log(lag(total_assets, 1))),
  total_debt_growth = 100 * (log(total_debt) - log(lag(total_debt, 1))),
  total_equity_growth = 100 * (log(total_equity) - log(lag(total_equity, 1))),
  repo_growth = 100 * (log(repo) - log(lag(repo, 1))),

  # growth variables in log form
  leverage_growth_log = log(leverage) - log(lag(leverage, 1)),
  total_assets_growth_log = log(total_assets) - log(lag(total_assets, 1)),
  total_equity_growth_log = log(total_equity) - log(lag(total_equity, 1)),
  total_repo_growth_log = log(repo) - log(lag(repo, 1))
) %>%
ungroup()

# these are the variables we want in the descriptive statistics
vars <- c(
  "leverage", "leverage_growth_calc",
  "var_99_gaussian", "var_99_gaussian_growth",
  "total_assets", "total_assets_growth",
  "total_debt", "total_debt_growth",
  "total_equity", "total_equity_growth",
  "repo", "repo_growth",
  "lcr", "cet1", "dp", "slr", "roa", "vix"
)

# scale some variables for readability
# this assumes balance sheet variables are in millions, so dividing by 1e3 gives usd billions
desc_data_scaled <- desc_data %>%
mutate(

```

```

    total_assets = total_assets / 1e3,
    total_debt = total_debt / 1e3,
    total_equity = total_equity / 1e3,
    repo = repo / 1e3
  )

# compute descriptive statistics
descriptive_stats <- desc_data_scaled %>%
  summarise(
    across(
      all_of(vars),
      list(
        N      = ~sum(!is.na(.x)),
        Mean   = ~mean(.x, na.rm = TRUE),
        SD     = ~sd(.x, na.rm = TRUE),
        Median = ~median(.x, na.rm = TRUE),
        Min    = ~min(.x, na.rm = TRUE),
        Max    = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  )

# reshape the output so it looks good in a table
descriptive_long <- descriptive_stats %>%
  pivot_longer(
    cols = everything(),
    names_to = c("variable", "statistic"),
    names_pattern = "^(.*)_(N|Mean|SD|Median|Min|Max)$",
    values_to = "value"
  ) %>%
  pivot_wider(
    names_from = statistic,
    values_from = value
  ) %>%
  mutate(

    # rename variables for presentation
    variable = dplyr::recode(
      variable,
      leverage = "Leverage",
      leverage_growth_calc = "Leverage Growth",
      var_99_gaussian = "Value-at-Risk",
      var_99_gaussian_growth = "Value-at-Risk Growth (%)",
      total_assets = "Total Assets (USD bn)",
      total_assets_growth = "Total Assets Growth (%)",
      total_debt = "Total Debt (USD bn)",
      total_debt_growth = "Total Debt Growth (%)",
      total_equity = "Equity (USD bn)",
      total_equity_growth = "Equity Growth (%)",
      repo = "Repo Funding (USD bn)",
      repo_growth = "Repo Funding Growth (%)",
      lcr = "Liquidity Coverage Ratio",
      cet1 = "CET1 Ratio",
      dp = "Dividend Payout Ratio",
      slr = "Supplementary Leverage Ratio",

```

```

    roa = "Return on Assets",
    vix = "VIX"
  ),

  # create sections in the table
  section = case_when(
    variable %in% c(
      "Leverage", "Total Assets (USD bn)",
      "Total Debt (USD bn)", "Equity (USD bn)",
      "Repo Funding (USD bn)"
    ) ~ "Balance Sheet Variables",
    variable %in% c(
      "Leverage Growth", "Total Assets Growth (%)",
      "Total Debt Growth (%)", "Equity Growth (%)",
      "Repo Funding Growth (%)"
    ) ~ "Growth Variables",
    variable %in% c(
      "Value-at-Risk", "Value-at-Risk Growth (%)"
    ) ~ "Risk Variables",
    variable %in% c(
      "Liquidity Coverage Ratio", "CET1 Ratio",
      "Supplementary Leverage Ratio"
    ) ~ "Regulatory Variables",
    variable %in% c(
      "Dividend Payout Ratio"
    ) ~ "Payout Policy",
    variable %in% c(
      "Return on Assets", "VIX"
    ) ~ "Control Variables"
  ),

  # set row order in the final table
  variable = factor(
    variable,
    levels = c(
      "Leverage",
      "Leverage Growth",
      "Total Assets (USD bn)",
      "Total Assets Growth (%)",
      "Total Debt (USD bn)",
      "Total Debt Growth (%)",
      "Equity (USD bn)",
      "Equity Growth (%)",
      "Repo Funding (USD bn)",
      "Repo Funding Growth (%)",
      "Value-at-Risk",
      "Value-at-Risk Growth (%)",
      "Liquidity Coverage Ratio",
      "CET1 Ratio",
      "Supplementary Leverage Ratio",
      "Dividend Payout Ratio",
      "Return on Assets",
      "VIX"
    )
  )
) %>%

```

```

arrange(variable)

# create the descriptive statistics table
descriptive_long %>%
  gt(rowname_col = "variable", groupname_col = "section") %>%
  cols_label(
    N = "N",
    Mean = "Mean",
    SD = "Std. Dev.",
    Median = "Median",
    Min = "Min",
    Max = "Max"
  ) %>%
  fmt_number(
    columns = c(Mean, SD, Median, Min, Max),
    decimals = 2
  ) %>%
  fmt_number(
    columns = N,
    decimals = 0
  ) %>%
  cols_align(
    align = "center",
    columns = c(N, Mean, SD, Median, Min, Max)
  ) %>%
  tab_header(
    title = "Descriptive Statistics",
    subtitle = "Balance sheet variables are reported in usd billions"
  ) %>%
  tab_options(
    table.font.size = px(13),
    data_row.padding = px(4),
    row_group.font.weight = "bold"
  )

```

| Descriptive Statistics                               |     |          |           |          |        |          |
|------------------------------------------------------|-----|----------|-----------|----------|--------|----------|
| Balance sheet variables are reported in usd billions |     |          |           |          |        |          |
|                                                      | N   | Mean     | Std. Dev. | Median   | Min    | Max      |
| <b>Balance Sheet Variables</b>                       |     |          |           |          |        |          |
| Leverage                                             | 512 | 10.76    | 1.49      | 10.66    | 7.24   | 15.19    |
| Total Assets (USD bn)                                | 512 | 1,517.09 | 998.49    | 1,505.78 | 153.97 | 4,560.20 |
| Total Debt (USD bn)                                  | 512 | 1,374.54 | 909.83    | 1,353.28 | 138.56 | 4,199.99 |
| Equity (USD bn)                                      | 512 | 142.55   | 91.86     | 124.35   | 15.41  | 362.44   |
| Repo Funding (USD bn)                                | 489 | 131.98   | 105.69    | 132.66   | 0.21   | 595.34   |
| <b>Growth Variables</b>                              |     |          |           |          |        |          |
| Leverage Growth                                      | 504 | 0.02     | 0.52      | 0.00     | -3.93  | 5.14     |
| Total Assets Growth (%)                              | 504 | 0.96     | 4.30      | 0.79     | -25.74 | 38.94    |
| Total Debt Growth (%)                                | 504 | 0.98     | 4.71      | 0.75     | -28.23 | 42.60    |

## Descriptive Statistics

Balance sheet variables are reported in usd billions

|                         | N   | Mean | Std. Dev. | Median | Min     | Max    |
|-------------------------|-----|------|-----------|--------|---------|--------|
| Equity Growth (%)       | 504 | 0.80 | 2.21      | 0.79   | -12.54  | 13.84  |
| Repo Funding Growth (%) | 481 | 0.15 | 32.36     | 0.18   | -244.57 | 221.90 |

### Risk Variables

|                          |     |       |       |       |        |        |
|--------------------------|-----|-------|-------|-------|--------|--------|
| Value-at-Risk            | 504 | 69.61 | 56.20 | 65.06 | 1.60   | 294.18 |
| Value-at-Risk Growth (%) | 496 | -1.17 | 23.43 | -1.51 | -97.23 | 154.69 |

### Regulatory Variables

|                              |     |        |       |        |      |        |
|------------------------------|-----|--------|-------|--------|------|--------|
| Liquidity Coverage Ratio     | 293 | 106.85 | 39.14 | 118.00 | 1.04 | 154.00 |
| CET1 Ratio                   | 393 | 12.74  | 1.68  | 12.39  | 9.80 | 17.70  |
| Supplementary Leverage Ratio | 366 | 6.66   | 1.00  | 6.60   | 4.00 | 9.51   |

### Payout Policy

|                       |     |       |        |       |      |          |
|-----------------------|-----|-------|--------|-------|------|----------|
| Dividend Payout Ratio | 496 | 39.63 | 228.89 | 26.06 | 0.00 | 4,990.48 |
|-----------------------|-----|-------|--------|-------|------|----------|

### Control Variables

|                  |     |       |      |       |       |       |
|------------------|-----|-------|------|-------|-------|-------|
| Return on Assets | 511 | 0.87  | 0.32 | 0.89  | -0.66 | 2.12  |
| VIX              | 384 | 17.98 | 5.28 | 16.52 | 10.31 | 34.49 |

```
desc_data <- desc_data %>%
  arrange(bank, desc(quarter))
```

More descriptive statistics

```

# bank group descriptive statistics

# this makes a copy so the original bank_data does not get changed
group_data <- bank_data %>%

  # make sure quarter is a date
  mutate(
    quarter = as.Date(quarter)
  ) %>%

  # sort inside this temporary dataset so lag works correctly
  arrange(bank, quarter) %>%

  # calculate variables within each bank
  group_by(bank) %>%
  mutate(

    # simple leverage measure
    leverage = total_assets / total_equity,

    # Leverage growth in percent
    leverage_growth_calc = 100 * (leverage / lag(leverage) - 1)
  ) %>%
  ungroup()

# here we create mean values by bank group
group_means <- group_data %>%
  group_by(bank_group) %>%
  summarise(
    n = n(),
    leverage = mean(leverage, na.rm = TRUE),
    leverage_growth = mean(leverage_growth_calc, na.rm = TRUE),
    var_99_gaussian = mean(var_99_gaussian, na.rm = TRUE),
    total_assets = mean(total_assets, na.rm = TRUE) / 1000,
    total_debt = mean(total_debt, na.rm = TRUE) / 1000,
    total_equity = mean(total_equity, na.rm = TRUE) / 1000,
    repo = mean(repo, na.rm = TRUE) / 1000,
    lcr = mean(lcr, na.rm = TRUE),
    cet1 = mean(cet1, na.rm = TRUE),
    slr = mean(slr, na.rm = TRUE),
    dp = mean(dp, na.rm = TRUE),
    roa = mean(roa, na.rm = TRUE),
    vix = mean(vix, na.rm = TRUE),
    .groups = "drop"
  )

# here we make the table look nicer
group_means_gt <- group_means %>%
  gt(rowname_col = "bank_group") %>%
  tab_header(
    title = "Average Values by Bank Group",
    subtitle = "Leverage growth is reported in percent. Assets, debt, equity, and repo are shown in usd billions."
  ) %>%
  cols_label(

```

```
n = "N",
leverage = "Leverage",
leverage_growth = "Leverage Growth (%)",
var_99_gaussian = "Value-at-Risk",
total_assets = "Total Assets",
total_debt = "Total Debt",
total_equity = "Equity",
repo = "Repo Funding",
lcr = "LCR",
cet1 = "CET1",
slr = "SLR",
dp = "Dividend Payout Ratio",
roa = "ROA",
vix = "VIX"
) %>%
fmt_number(
  columns = c(
    leverage, leverage_growth, var_99_gaussian, total_assets,
    total_debt, total_equity, repo, lcr, cet1, slr, dp, roa, vix
  ),
  decimals = 2
) %>%
tab_options(
  table.font.size = px(13),
  heading.title.font.size = px(16),
  heading.subtitle.font.size = px(12)
)
```

group\_means\_gt

Average Values by Bank Group

Leverage growth is reported in percent. Assets, debt, equity, and repo are shown in usd billions.

|           | N   | Leverage | Leverage<br>Growth<br>(%) | Value-<br>at-<br>Risk | Total<br>Assets | Total<br>Debt | Equity | Repo<br>Funding | LCR    | CET1  | SLR  | Dividend<br>Payout<br>Ratio | R |
|-----------|-----|----------|---------------------------|-----------------------|-----------------|---------------|--------|-----------------|--------|-------|------|-----------------------------|---|
| custodian | 128 | 10.53    | 0.80                      | 7.24                  | 323.90          | 292.53        | 31.37  | 8.69            | 61.79  | 11.90 | 6.34 | 32.00                       | C |
| market    | 128 | 11.75    | 0.08                      | 104.86                | 1,053.14        | 964.23        | 88.91  | 122.85          | 131.01 | 14.52 | 6.69 | 36.72                       | C |
| universal | 256 | 10.37    | 0.09                      | 83.16                 | 2,345.65        | 2,120.71      | 224.95 | 204.73          | 117.68 | 12.25 | 6.82 | 45.00                       | C |



G-sib plot;

```

gsib <-
  read_excel("C:\\Users\\Eier\\Desktop\\Master balance sheet\\gsib.xlsx")

gsiboverall <- gsib %>%
  filter(id == "overall") %>%
  pivot_longer(
    cols = -c(year, id),
    names_to = "bank",
    values_to = "score"
  )

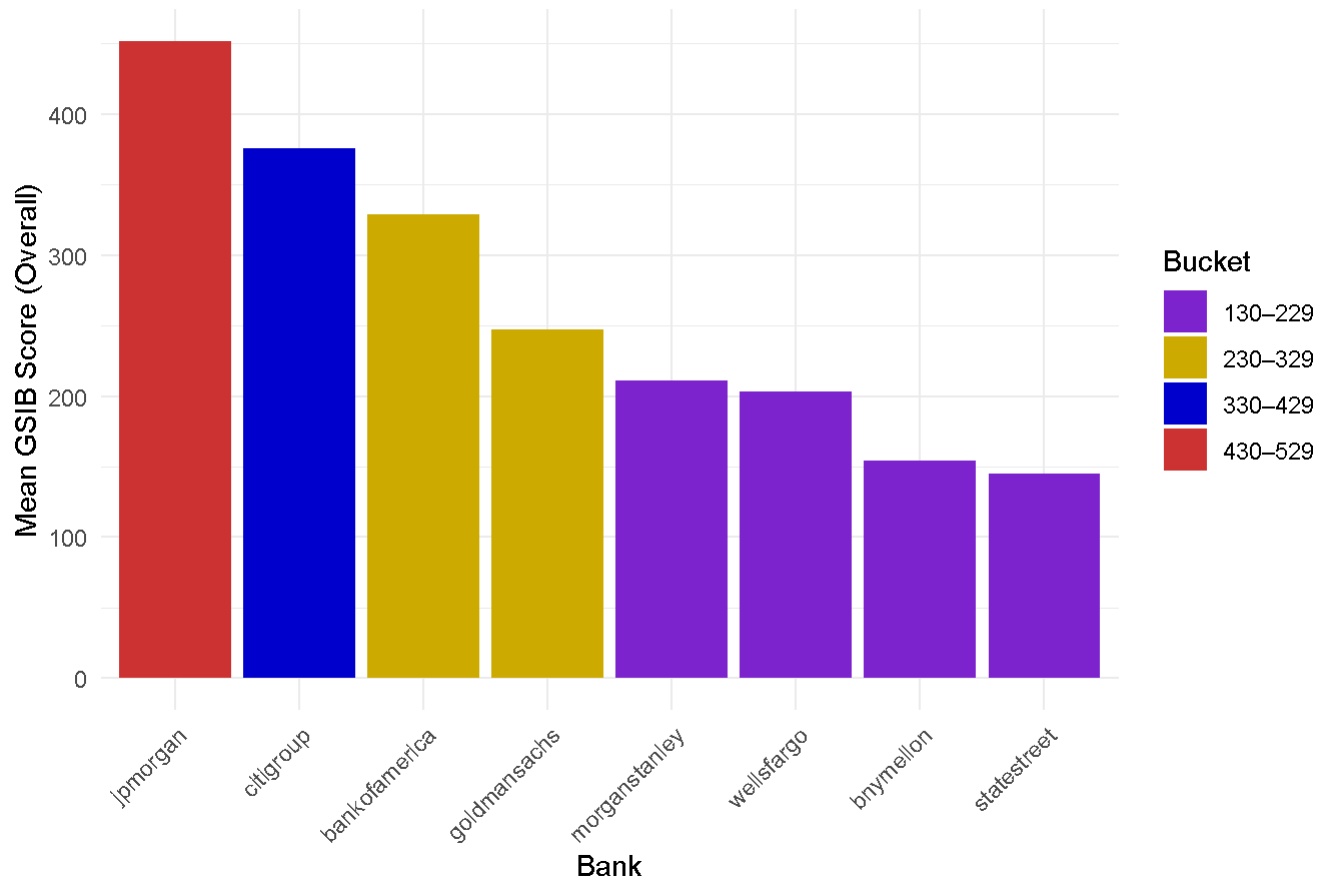
gsiboverall <- gsiboverall |>
  group_by(bank) |>
  summarise(mean_score = round(mean(score, na.rm = TRUE))) |>
mutate(
  bucket = case_when(
    mean_score >= 430 & mean_score <= 529 ~ "430-529",
    mean_score >= 330 & mean_score <= 429 ~ "330-429",
    mean_score >= 230 & mean_score <= 329 ~ "230-329",
    mean_score >= 130 & mean_score <= 229 ~ "130-229"
  ),
  bank = reorder(bank, desc(mean_score)))

ggplot(gsiboverall, aes(x = bank, y = mean_score, fill = bucket)) +
  geom_col() +
  scale_fill_manual(
    values = c(
      "430-529" = "brown3",
      "330-429" = "blue3",
      "230-329" = "gold3",
      "130-229" = "purple3"
    )
  ) +
  labs(
    title = "Average GSIB Overall Scores by Bank",
    x = "Bank",
    y = "Mean GSIB Score (Overall)",
    fill = "Bucket"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      box.color = "black",
      box.background = "white",
      margin = margin(b = 10)
    )
  )

```



### Average GSIB Overall Scores by Bank



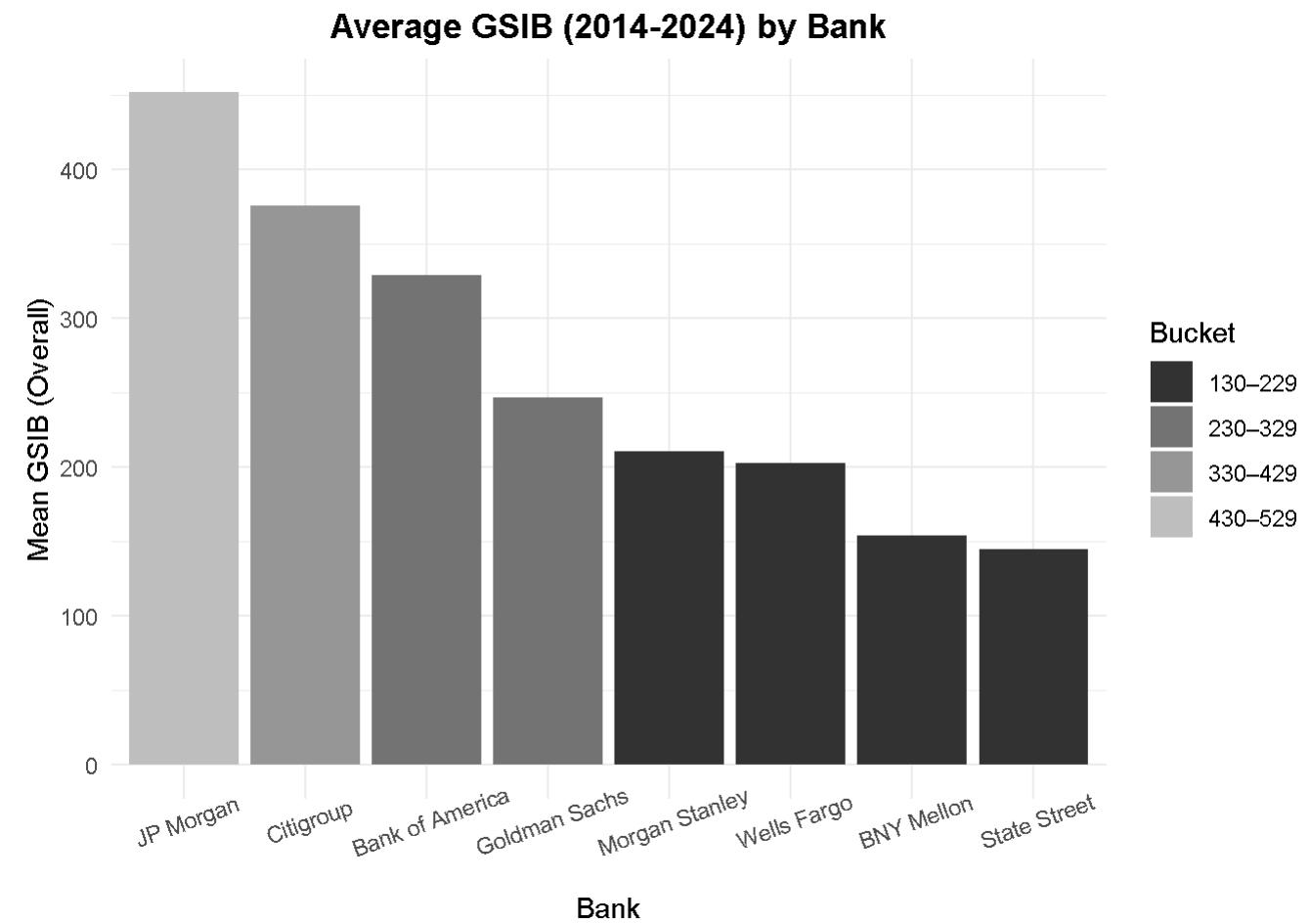
```

gsib_mean <- gsib %>%
  filter(id == "overall") %>%
  pivot_longer(
    cols = -c(year, id),
    names_to = "bank",
    values_to = "score"
  ) %>%
  group_by(bank) %>%
  summarise(mean_score = round(mean(score, na.rm = TRUE))) %>%
  mutate(
    bucket = case_when(
      mean_score >= 430 & mean_score <= 529 ~ "430-529",
      mean_score >= 330 & mean_score <= 429 ~ "330-429",
      mean_score >= 230 & mean_score <= 329 ~ "230-329",
      mean_score >= 130 & mean_score <= 229 ~ "130-229"
    ),
    bank = reorder(bank, desc(mean_score))
  )

ggplot(gsiboverall, aes(x = bank, y = mean_score, fill = bucket)) +
  geom_col() +
  scale_fill_manual(
    values = c(
      "430-529" = "grey",
      "330-429" = "grey60",
      "230-329" = "grey45",
      "130-229" = "grey20"
    )
  ) + labs(
    title = "Average GSIB (2014-2024) by Bank",
    x = "Bank",
    y = "Mean GSIB (Overall)",
    fill = "Bucket"
  ) +
  scale_x_discrete(
    labels = c(
      "jpmorgan" = "JP Morgan",
      "bankofamerica" = "Bank of America",
      "citigroup" = "Citigroup",
      "goldmansachs" = "Goldman Sachs",
      "morganstanley" = "Morgan Stanley",
      "wellsfargo" = "Wells Fargo",
      "bnymellon" = "BNY Mellon",
      "statestreet" = "State Street"
    )
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 20
    ),
    plot.title = element_text(
      hjust = 0.5,

```

```
face = "bold"
))
```



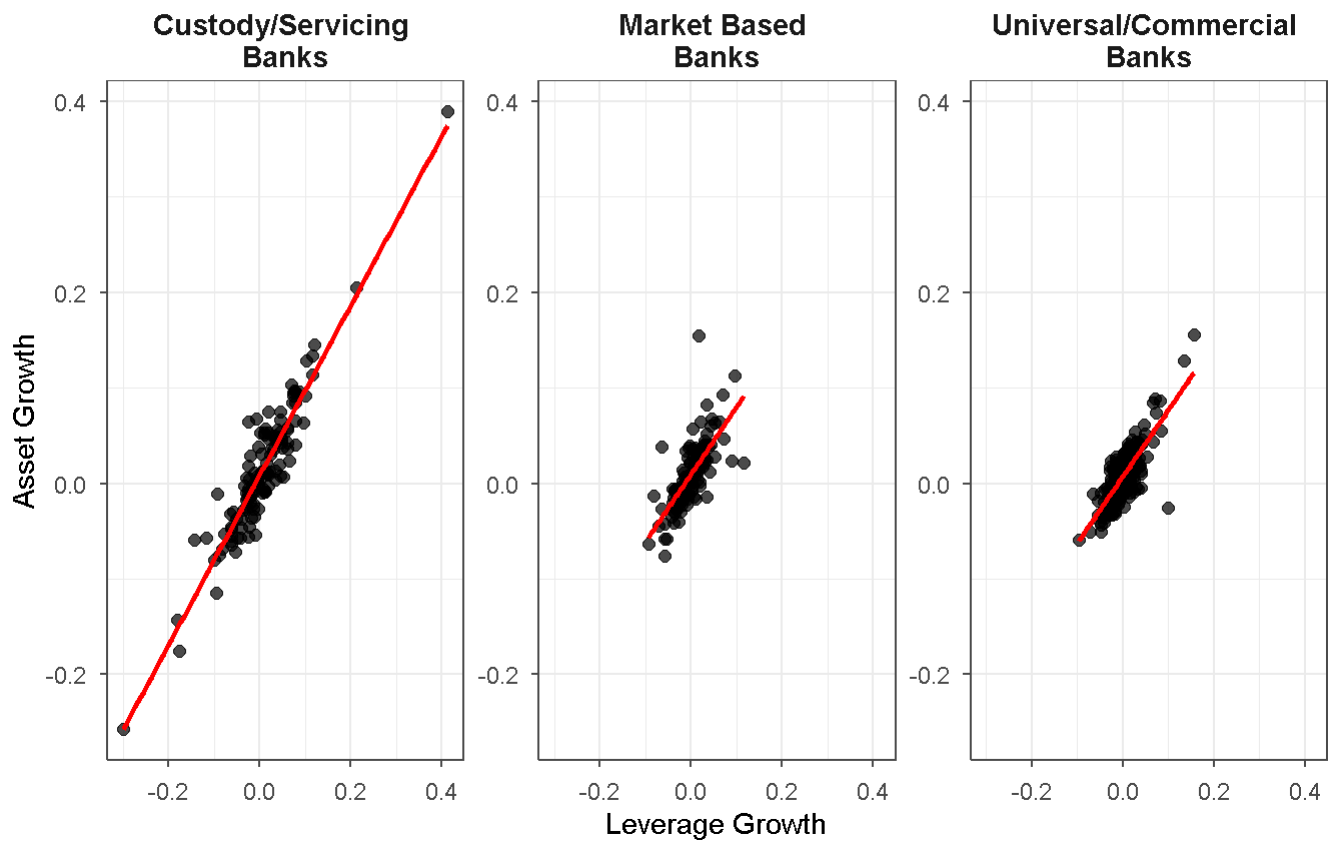
Growth figures

```
# using the descriptive statistic dataset

ggplot(desc_data, aes(x = leverage_growth_log, y = total_assets_growth_log)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap2(
    ~ bank_group,
    nrow = 1,
    axes = "all",
    labeller = as_labeller(c(
      "custodian" = "Custody/Servicing \nBanks",
      "market" = "Market Based \nBanks",
      "universal" = "Universal/Commercial \nBanks"
    ))
  ) +
  labs(
    title = "Correlation Between Asset Growth and Leverage Growth",
    subtitle = "(Log Form)",
    x = "Leverage Growth",
    y = "Asset Growth"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.line = element_line(color = "grey30"),
    axis.ticks = element_line(color = "grey30"),
    panel.border = element_rect(color = "grey30", fill = NA),
    strip.text = element_text(size = 11, face = "bold")
  )
```

## Correlation Between Asset Growth and Leverage Growth

(Log Form)



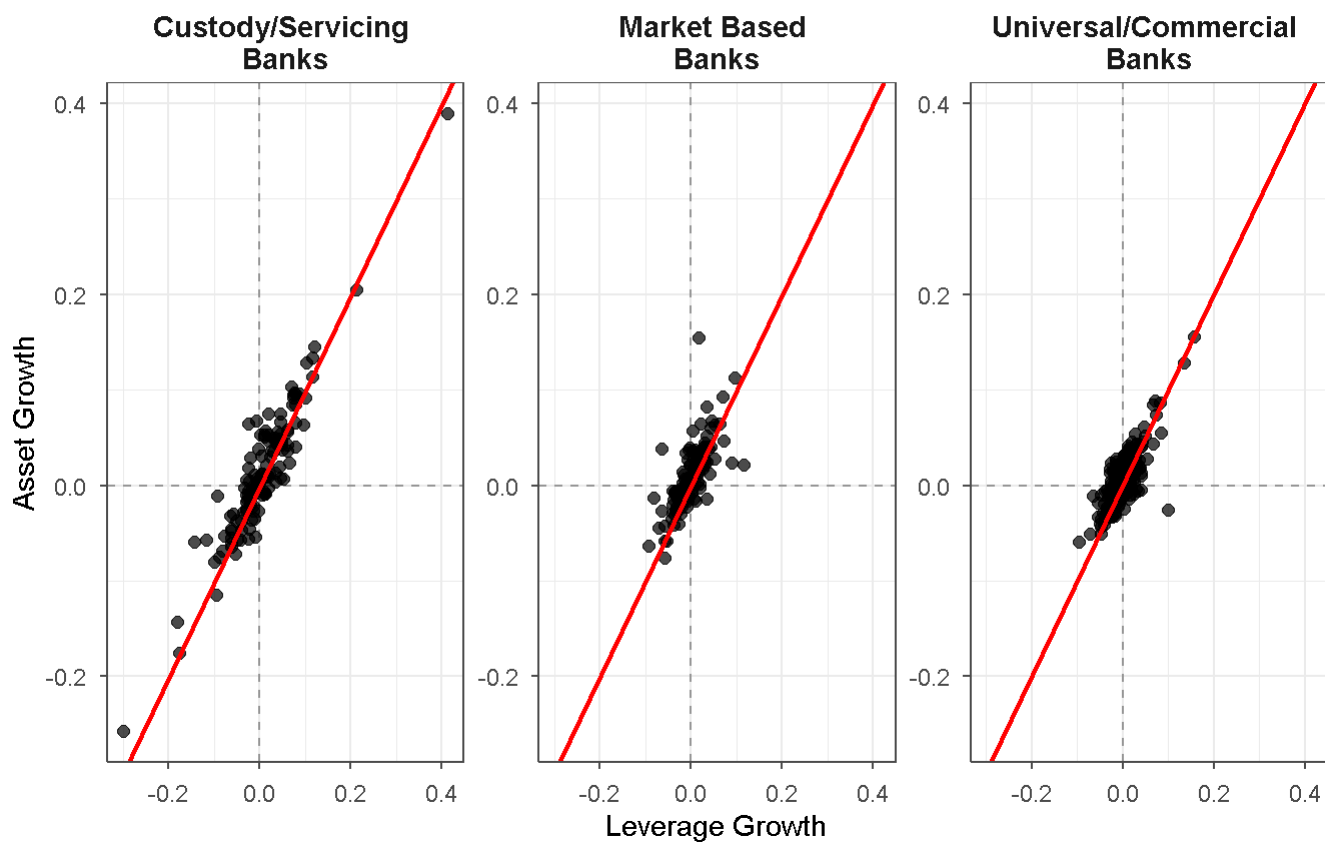
```

ggplot(desc_data, aes(x = leverage_growth_log, y = total_assets_growth_log)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey60", linewidth = 0.5) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey60", linewidth = 0.5) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 0, slope = 1, color = "red", linewidth = 1) +
  facet_wrap2(
    ~ bank_group,
    nrow = 1,
    axes = "all",
    labeller = as_labeller(c(
      "custodian" = "Custody/Servicing \nBanks",
      "market" = "Market Based \nBanks",
      "universal" = "Universal/Commercial \nBanks"
    ))
  ) +
  labs(
    title = "Asset Growth and Leverage Growth",
    subtitle = "(Log Form)",
    x = "Leverage Growth",
    y = "Asset Growth"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.line = element_line(color = "grey30"),
    axis.ticks = element_line(color = "grey30"),
    panel.border = element_rect(color = "grey30", fill = NA),
    strip.text = element_text(size = 11, face = "bold")
  )

```

## Asset Growth and Leverage Growth

(Log Form)



```

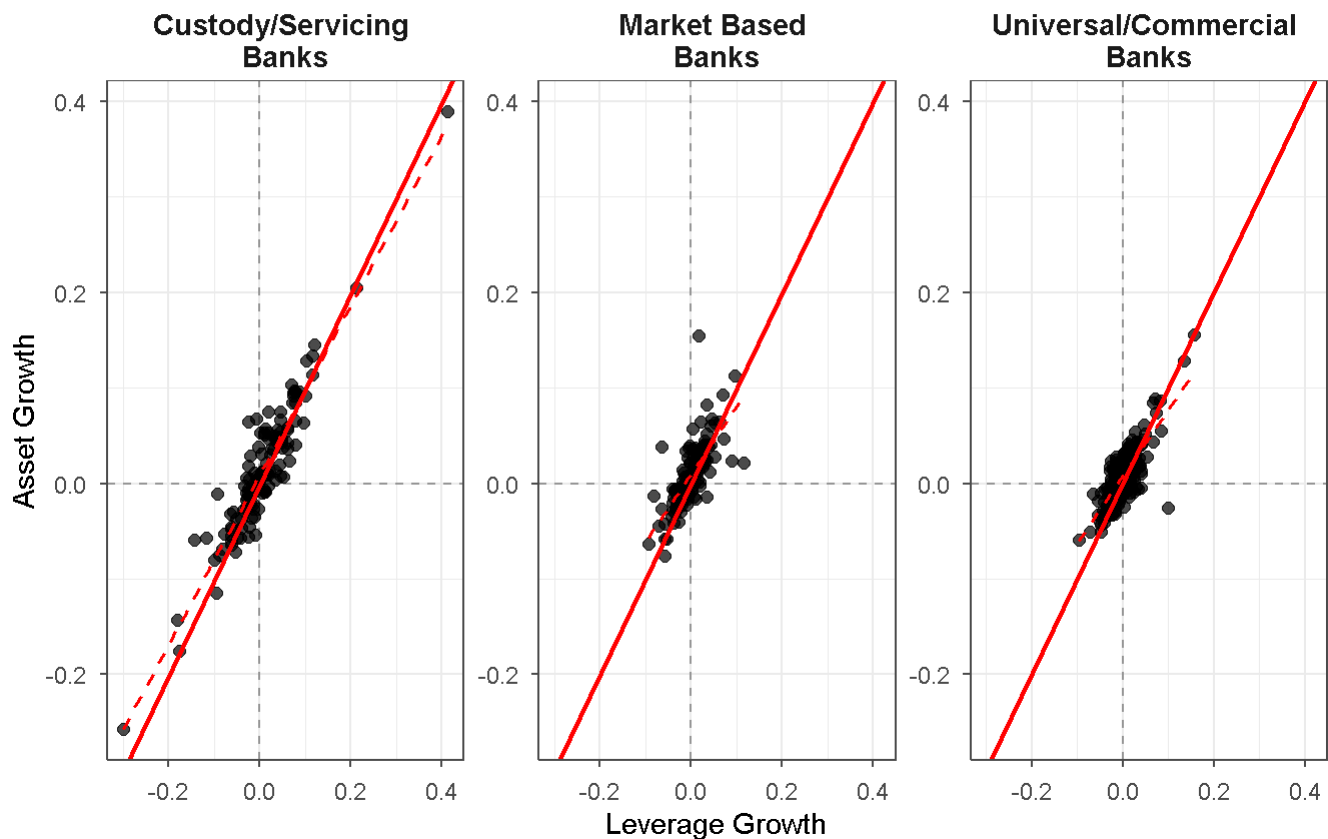
ggplot(desc_data, aes(x = leverage_growth_log, y = total_assets_growth_log)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey60", linewidth = 0.5) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey60", linewidth = 0.5) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "red", linetype = "dashed", linewidth = 0.7, a
lpha = 0.6) +
  geom_abline(intercept = 0, slope = 1, color = "red", linewidth = 1) +
  facet_wrap2(
    ~ bank_group,
    nrow = 1,
    axes = "all",
    labeller = as_labeller(c(
      "custodian" = "Custody/Servicing \nBanks",
      "market" = "Market Based \nBanks",
      "universal" = "Universal/Commercial \nBanks"
    ))
  ) +
  labs(
    title = "Asset Growth and Leverage Growth",
    subtitle = "(Log Form)",
    x = "Leverage Growth",
    y = "Asset Growth"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.line = element_line(color = "grey30"),
    axis.ticks = element_line(color = "grey30"),
    panel.border = element_rect(color = "grey30", fill = NA),
    strip.text = element_text(size = 11, face = "bold")
  )

```



## Asset Growth and Leverage Growth

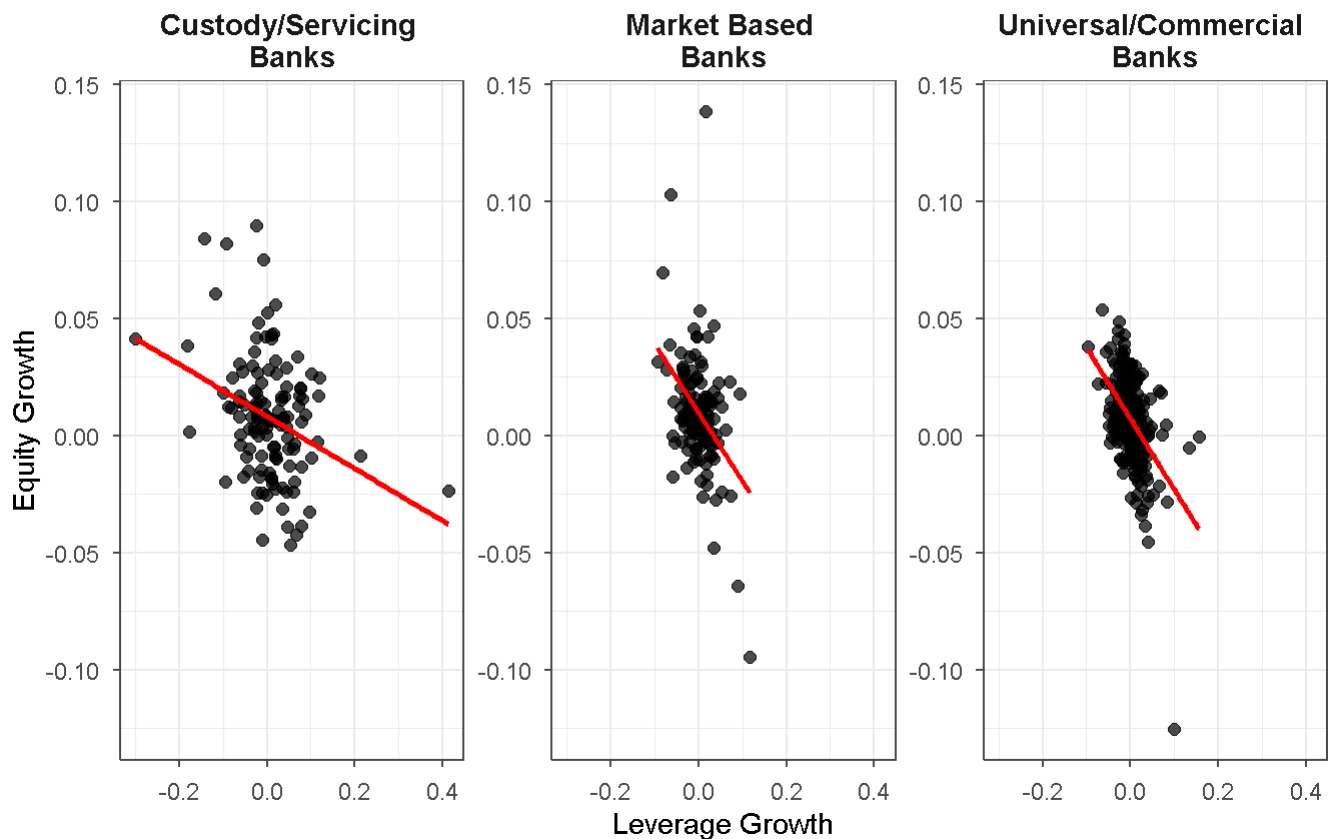
(Log Form)



```
ggplot(desc_data, aes(x = leverage_growth_log, y = total_equity_growth_log)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap2(
    ~ bank_group,
    nrow = 1,
    axes = "all",
    labeller = as_labeller(c(
      "custodian" = "Custody/Servicing \nBanks",
      "market" = "Market Based \nBanks",
      "universal" = "Universal/Commercial \nBanks"
    ))
  ) +
  labs(
    title = "Correlation Between Equity Growth and Leverage Growth",
    subtitle = "(Log Form)",
    x = "Leverage Growth",
    y = "Equity Growth"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.line = element_line(color = "grey30"),
    axis.ticks = element_line(color = "grey30"),
    panel.border = element_rect(color = "grey30", fill = NA),
    strip.text = element_text(size = 11, face = "bold")
  )
```

## Correlation Between Equity Growth and Leverage Growth

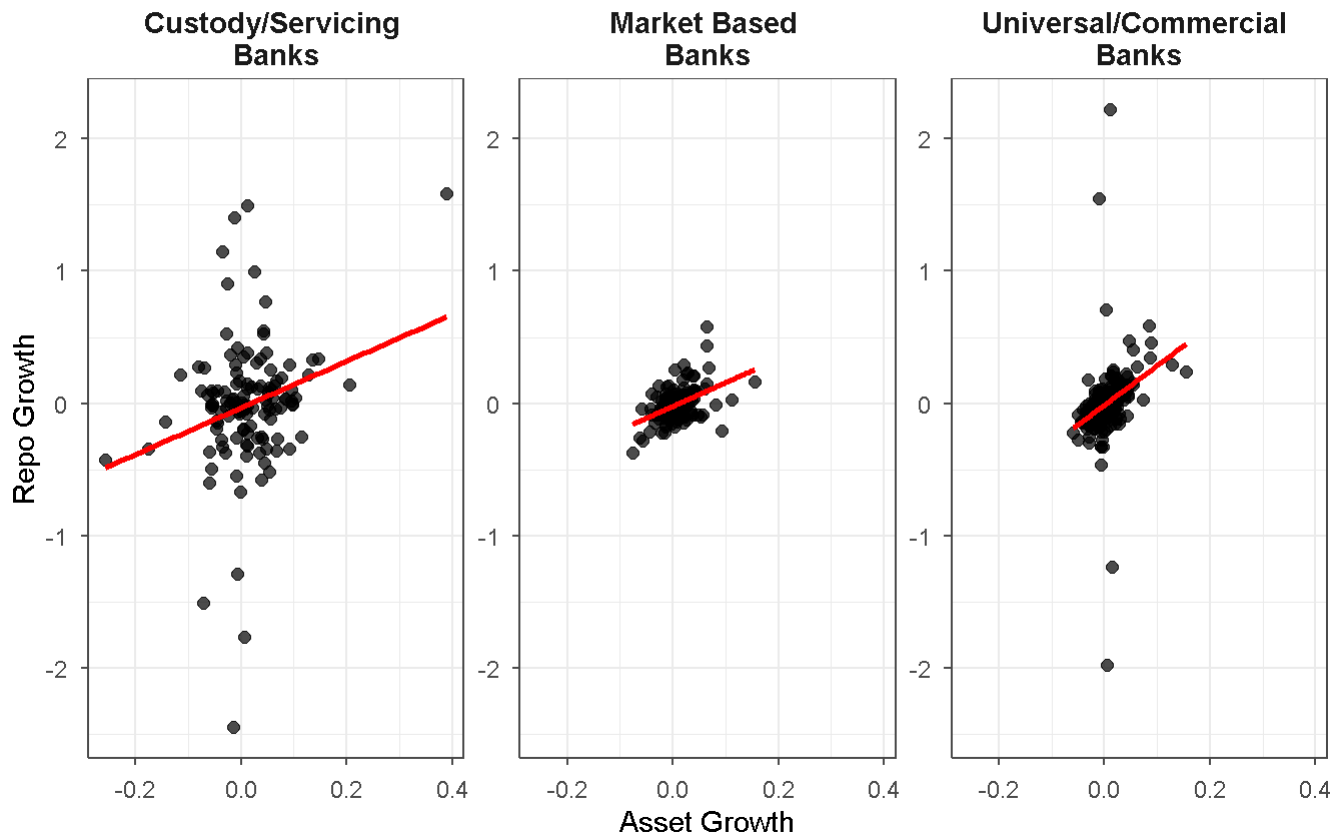
(Log Form)



```
ggplot(desc_data, aes(x = total_assets_growth_log, y = total_repo_growth_log)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap2(
    ~ bank_group,
    nrow = 1,
    axes = "all",
    labeller = as_labeller(c(
      "custodian" = "Custody/Servicing \nBanks",
      "market" = "Market Based \nBanks",
      "universal" = "Universal/Commercial \nBanks"
    ))
  ) +
  labs(
    title = "Correlation Between Repo Growth and Asset Growth",
    subtitle = "(Log Form)",
    x = "Asset Growth",
    y = "Repo Growth"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.line = element_line(color = "grey30"),
    axis.ticks = element_line(color = "grey30"),
    panel.border = element_rect(color = "grey30", fill = NA),
    strip.text = element_text(size = 11, face = "bold")
  )
```

## Correlation Between Repo Growth and Asset Growth

(Log Form)



```

# making a text column for showing quarters in the graph
desc_data <- desc_data %>%
  mutate(quarter_label = as.character(quarter))

# chosing the most extreme values
label_data <- desc_data %>%
  filter(!is.na(total_assets_growth_log),
         !is.na(total_repo_growth_log),
         !is.na(quarter_label)) %>%
  group_by(bank_group) %>%
  mutate(outlier_score = abs(total_repo_growth_log)) %>%
  slice_max(order_by = outlier_score, n = 4, with_ties = FALSE) %>%
  ungroup()

ggplot(desc_data, aes(x = total_assets_growth_log, y = total_repo_growth_log)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  geom_text_repel(
    data = label_data,
    mapping = aes(
      x = total_assets_growth_log,
      y = total_repo_growth_log,
      label = quarter_label
    ),
    inherit.aes = FALSE,
    size = 3.5,
    colour = "black",
    min.segment.length = 0,
    box.padding = 0.35,
    point.padding = 0.25,
    max.overlaps = Inf,
    na.rm = TRUE
  ) +
  facet_wrap2(
    ~ bank_group,
    nrow = 1,
    axes = "all",
    labeller = as_labeller(c(
      "custodian" = "Custody/Servicing \nBanks",
      "market" = "Market Based \nBanks",
      "universal" = "Universal/Commercial \nBanks"
    ))
  ) +
  labs(
    title = "Correlation Between Repo Growth and Asset Growth",
    subtitle = "(Log Form)",
    x = "Asset Growth",
    y = "Repo Growth"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.line = element_line(color = "grey30"),
    axis.ticks = element_line(color = "grey30"),

```

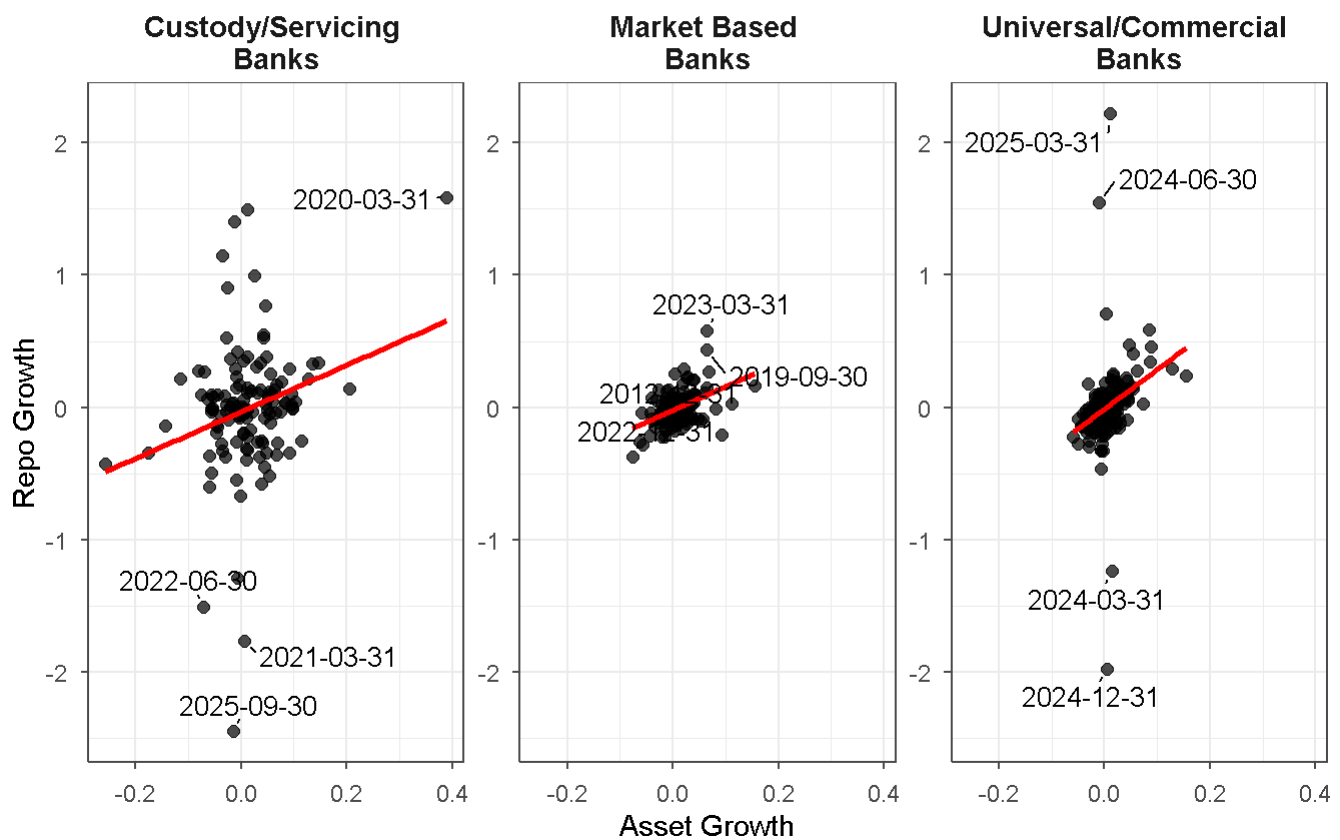
```

panel.border = element_rect(color = "grey30", fill = NA),
strip.text = element_text(size = 11, face = "bold")
)

```

## Correlation Between Repo Growth and Asset Growth

(Log Form)



dividend payouts and equity

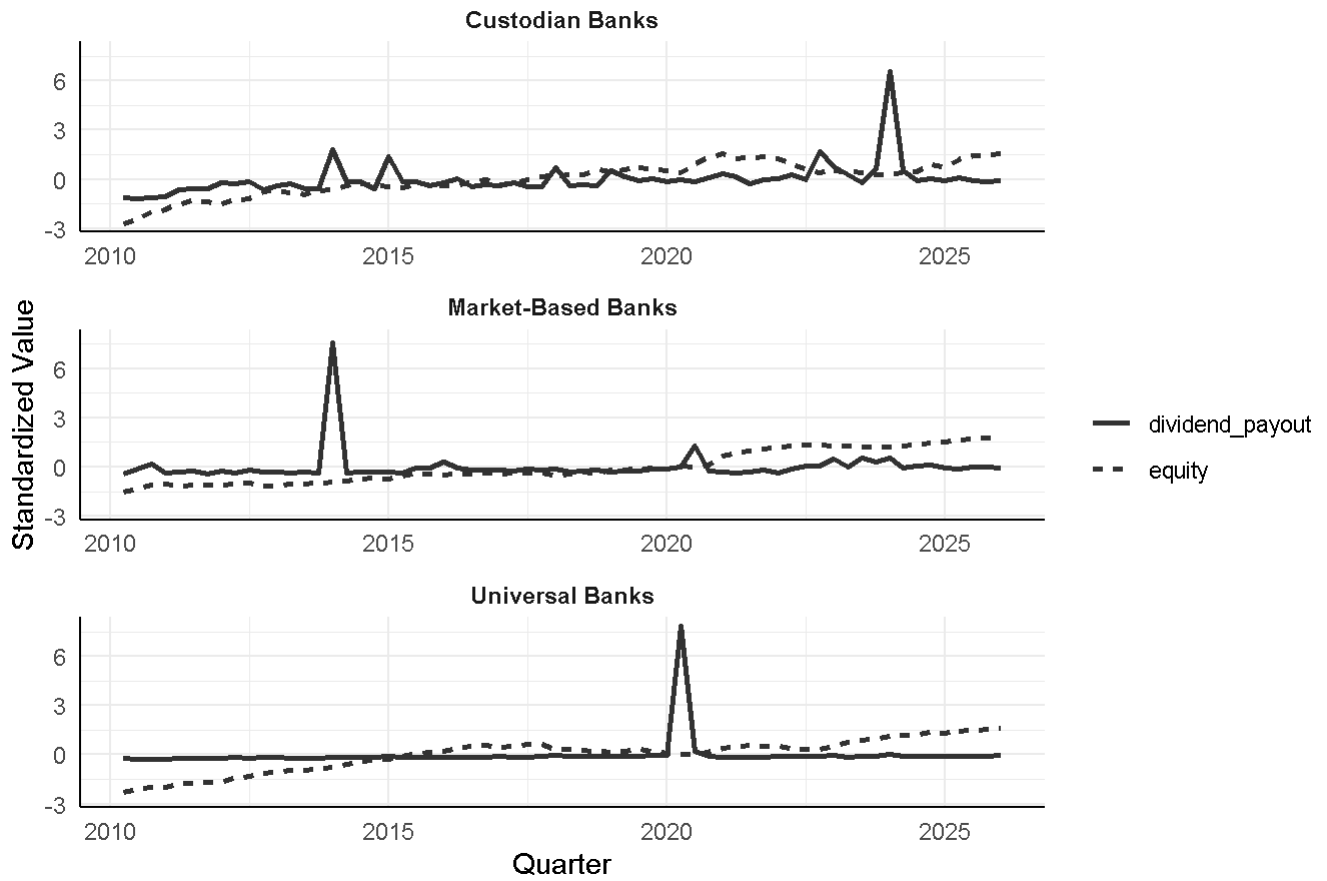
```

plot_data_equity <- desc_data %>%
  group_by(bank_group, quarter) %>%
  summarise(
    dividend_payout = mean(dp, na.rm = TRUE),
    equity = mean(total_equity, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(bank_group, quarter) %>%
  group_by(bank_group) %>%
  mutate(
    dividend_payout = as.numeric(scale(dividend_payout)),
    equity = as.numeric(scale(equity))
  ) %>%
  ungroup() %>%
  pivot_longer(
    cols = c(dividend_payout, equity),
    names_to = "Variable",
    values_to = "Value"
  )

ggplot(plot_data_equity, aes(x = quarter, y = Value, group = Variable, linetype = Variable)) +
  geom_line(linewidth = 1, color = "grey20") +
  facet_wrap2(
    ~ bank_group,
    ncol = 1,
    axes = "all",
    labeller = as_labeller(c(
      "custodian" = "Custodian Banks",
      "market" = "Market-Based Banks",
      "universal" = "Universal Banks"
    ))
  ) +
  labs(
    title = "Dividend Payout and Equity Over Time by Bank Group",
    x = "Quarter",
    y = "Standardized Value",
    linetype = ""
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.line.x = element_line(color = "black"),
    axis.line.y = element_line(color = "black"),
    strip.text = element_text(face = "bold")
  )

```

## Dividend Payout and Equity Over Time by Bank Group



Repo graph

```

# Custodian
plot_custodian <- desc_data %>%
  filter(bank_group == "custodian") %>%
  group_by(quarter) %>%
  summarise(
    repo = mean(repo, na.rm = TRUE),
    leverage = mean(leverage, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(quarter) %>%
  mutate(
    repo = as.numeric(scale(repo)),
    leverage = as.numeric(scale(leverage))
  ) %>%
  pivot_longer(
    cols = c(repo, leverage),
    names_to = "Variable",
    values_to = "Value"
  ) %>%
  ggplot(aes(x = quarter, y = Value, group = Variable, linetype = Variable)) +
  geom_line(linewidth = 1, color = "grey20") +
  labs(
    title = "Custodian Banks",
    x = "Quarter",
    y = "Standardized Value",
    linetype = ""
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "black"),
    axis.line.x = element_line(color = "black"),
    axis.line.y = element_line(color = "black"),
    axis.ticks = element_line(color = "black"),
    panel.border = element_rect(color = "black", fill = NA),
    legend.position = "bottom"
  )

# Market
plot_market <- desc_data %>%
  filter(bank_group == "market") %>%
  group_by(quarter) %>%
  summarise(
    repo = mean(repo, na.rm = TRUE),
    leverage = mean(leverage, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(quarter) %>%
  mutate(
    repo = as.numeric(scale(repo)),
    leverage = as.numeric(scale(leverage))
  ) %>%
  pivot_longer(
    cols = c(repo, leverage),

```



```

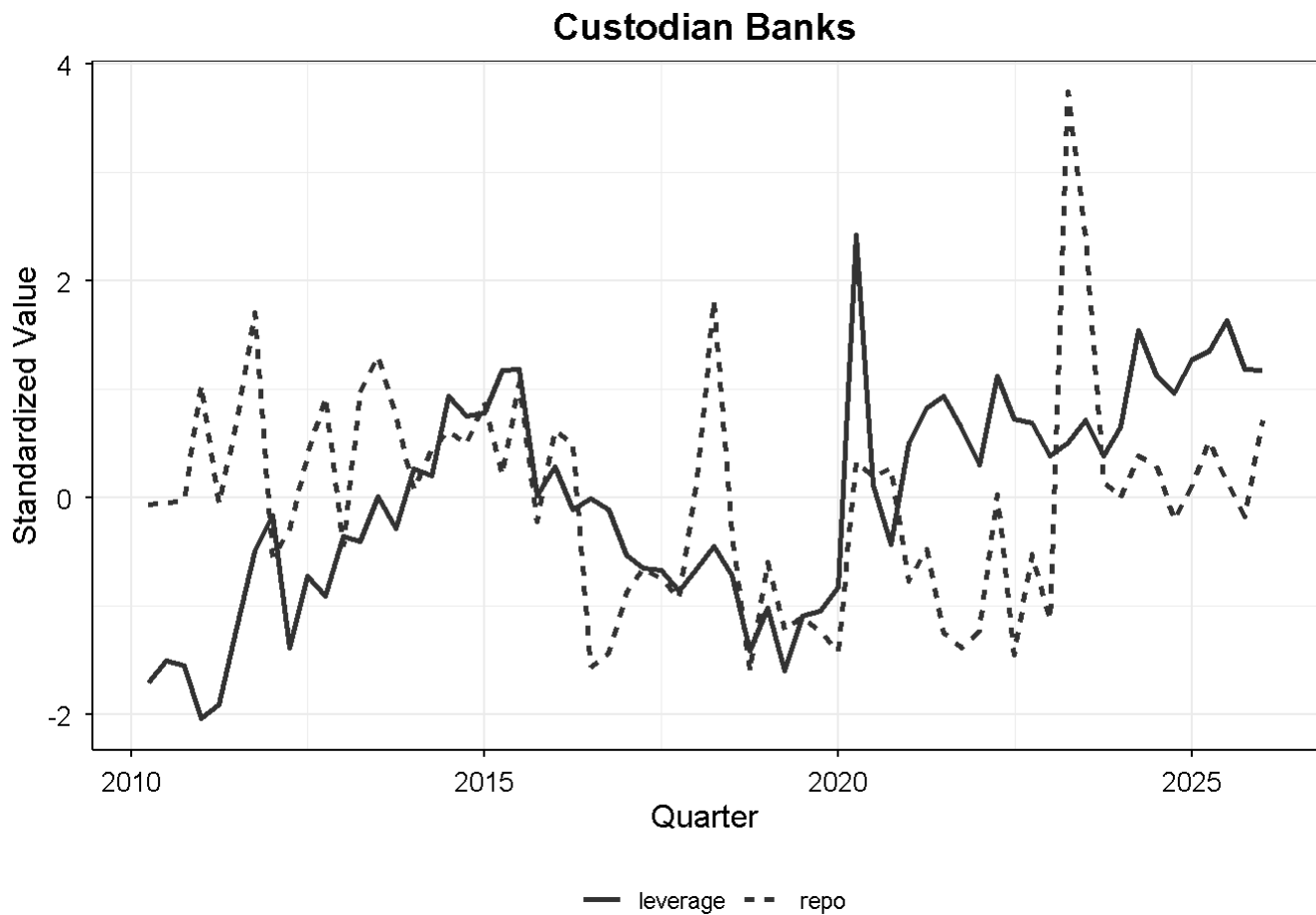
    names_to = "Variable",
    values_to = "Value"
  ) %>%
  ggplot(aes(x = quarter, y = Value, group = Variable, linetype = Variable)) +
  geom_line(linewidth = 1, color = "grey20") +
  labs(
    title = "Market-Based Banks",
    x = "Quarter",
    y = "Standardized Value",
    linetype = ""
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "black"),
    axis.line.x = element_line(color = "black"),
    axis.line.y = element_line(color = "black"),
    axis.ticks = element_line(color = "black"),
    panel.border = element_rect(color = "black", fill = NA),
    legend.position = "bottom"
  )

# Universal
plot_universal <- desc_data %>%
  filter(bank_group == "universal") %>%
  group_by(quarter) %>%
  summarise(
    repo = mean(repo, na.rm = TRUE),
    leverage = mean(leverage, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(quarter) %>%
  mutate(
    repo = as.numeric(scale(repo)),
    leverage = as.numeric(scale(leverage))
  ) %>%
  pivot_longer(
    cols = c(repo, leverage),
    names_to = "Variable",
    values_to = "Value"
  ) %>%
  ggplot(aes(x = quarter, y = Value, group = Variable, linetype = Variable)) +
  geom_line(linewidth = 1, color = "grey20") +
  labs(
    title = "Universal Banks",
    x = "Quarter",
    y = "Standardized Value",
    linetype = ""
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "black"),
    axis.line.x = element_line(color = "black"),

```

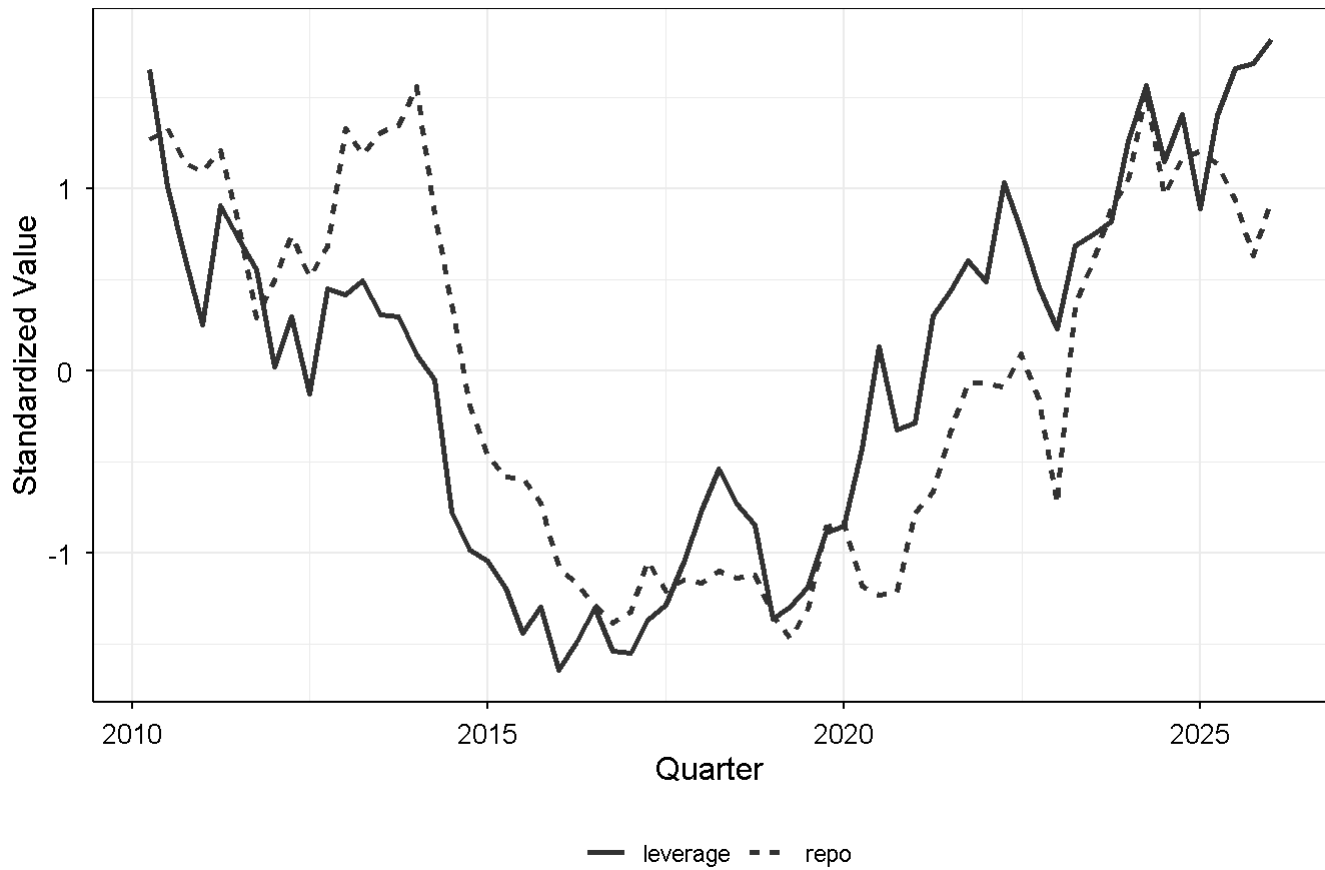
```
axis.line.y = element_line(color = "black"),  
axis.ticks = element_line(color = "black"),  
panel.border = element_rect(color = "black", fill = NA),  
legend.position = "bottom"  
)
```

plot\_custodian



plot\_market

## Market-Based Banks



```
plot_universal
```

## Universal Banks

